

		Total number of fringes = $3 + 1 + 3 = 7$
17	B	Using Rayleigh's criterion for resolution: $\theta = \frac{\lambda}{b}$ For small angle, $\theta = \sin \theta = \tan \theta$ $(700 \times 10^{-9}) / (30 \times 10^{-3}) = x / 5.0$ $x = 1.2 \times 10^{-3} \text{ m} = 0.12 \text{ mm}$
18	B	The electric field lines must always be perpendicular to the surfaces of the charged